

EAP-Architecture Specification and Architecture

A7: Web Resources Specification

An artifact that seeks to resolve into discrete sections the whole context and implementation of the BountyBust website. For that, we define modules, privilege instances, and existing web APIs described with OpenAPI.

1. Module Overview

M01: Account creation and accessibility	Content related to methods of account accessibility and creation, as well as a personal space that features its details and platform actions taken by the user. Features include login/logout, registration, password recovery, viewing and editing personal profiles, and viewing users' created content.
M02: Bounties	Content based on platform questions designated as Bounties. Features include following, creating, editing, and deleting a bounty, viewing a bounty's details, and bounty rating.
M03: Answers	Content inserted in the scope of Bounty's answers. Features include posting, editing, and deleting an answer, answer rating, and a visual indication if marked as correct.
M04: Comments	Describes a form of interaction with the two previously described pieces of content. Features include posting, editing, and deleting a comment, and comment rating (upvote only for exposure, doesn't need to add qualitative value to the context).
M05: Tags	Content that introduces a form of labelling Bounties amongst different topics available on the platform. Features include following, creating, editing, and deleting tags (*privilege-based*) and viewing their details.
M06: Rating and award system	Content related to features that are intended to give value and promote users' actions and content in the platform. Features include upvoting and downvoting questions and answers, upvoting comments, a user and bounty reward system, a leaderboard, and an achievement/badge system.
M07: Search and filters	Content related to searching and filtering features on the platform's content. Features include full text and exact match search across questions and answers, filtering and ordering by creation dates, ratings, tags, amongst others...

M08: Platform regulation and information pages	Content related to platform privilege hierarchy functionalities and static pages containing extra information about the platform and its creators. Features include, as administrators, viewing, searching, creating, deleting, and banning a user's account, electing new administrators, granting or revoking moderator privileges, adding or removing tags, and managing account appeals. As moderators, delete unsuitable content, edit misplaced tags from bounties, and accept appeals for deleted bounties and answers.
M09: Notification system	Web resources associated with the BountyBust platform and personal notifications. Features include notifications in replying and rating to content, viewing the notification list, viewing personal notifications separately, and marking notifications as read.

2. Permissions

PUB	Public	Platform visitor or unauthenticated user with no permissions
USR	User	Platform authenticated user
OWN	Owner	User responsible for the creation of an item (bounty, answer, content)
MOD	Moderator	User with moderator's privileges in BountyBust
ADM	Administrator	BountyBust's platform administrators

3. OpenAPI Specification

A small section of the OpenAPI specification file is provided below; you can find the whole transcript at the following link:

https://gitlab.up.pt/lbaw/lbaw2526/lbaw25146/-/blob/main/EAP/a7_openapi.yaml

```
openapi: 3.0.0
info:
  version: 1.0.0
  title: "BountyBust Web API"
externalDocs:
  description: More info can be found here.
```

14/11/25

```
url: https://gitlab.up.pt/lbaw/lbaw2526/lbaw25146
servers:
  - url: http://lbaw.fe.up.pt
    description: Production server
tags:
  - name: "M01: Account creation and accessibility"
  - name: "M02: Bounties"
  - name: "M03: Answers"
  - name: "M04: Comments"
  - name: "M05: Tags"
  - name: "M06: Rating and award system"
  - name: "M07: Search and filters"
  - name: "M08: Platform regulation and information pages"
  - name: "M09: Notification system"
paths:
  /login:
    get:
      operationId: R101
      summary: "R101: Login Form"
      description: "Get to the login form page. Access: PUB"
      tags:
        - "M01: Account creation and accessibility"
      responses:
        "200":
          description: "Ok. Login Form"
    post:
      operationId: R102
      summary: "R102: Login Action"
      description: "Check submitted user credentials from Login form. Access:
PUB"
      tags:
        - "M01: Account creation and accessibility"
      requestBody:
        required: true
        content:
          application/x-www-form-urlencoded:
            schema:
              type: object
              properties:
                username:
                  type: string
                password:
                  type: string
              required:
                - username
                - password
      responses:
```

```

    "302":
      description: "Redirect after checking user credentials."
      headers:
        Location:
          schema:
            type: string
          examples:
            302Success:
              description: "Correct credentials, redirecting to profile."
              value: "/users/{id}"
            302Error:
              description: "User failed to connect, redirecting to login
form."
              value: "/login"
/logout:
  post:
    operationId: R103
    summary: "R103: Logout Action"
    description: "Logout of the currently authenticated account. Access:
USR,MOD,ADM"
    tags:
      - "M01: Account creation and accessibility"
    responses:
      "302":
        description: "Redirect after logout"
        headers:
          Location:
            schema:
              type: string
            examples:
              302Success:
                description: "Successfully logging out, redirecting to
homepage"
                value: "/"
/register:
  get:
    operationId: R104
    summary: "R104: Register Form"
    description: "Get register form page. Access: PUB"
    tags:
      - "M01: Account creation and accessibility"
    responses:
      "200":
        description: "Ok. Register Form"
  post:
    operationId: R105
    summary: "R105: Register Action"
    description: "Handle new user account creation form. Access: PUB"

```

```

tags:
  - "M01: Account creation and accessibility"
requestBody:
  required: true
  content:
    multipart/form-data:
      schema:
        type: object
        properties:
          username:
            type: string
          email:
            type: string
          password:
            type: string
          name:
            type: string
          profile_picture:
            type: string
            format: binary
        required:
          - username
          - email
          - password
          - name
responses:
  "302":
    description: "Redirect after registering."
    headers:
      Location:
        schema:
          type: string
        examples:
          302Success:
            description: "Successfully authenticated, redirecting to
user profile."
            value: "/users/{id}"
          302Failure:
            description: "Failed to authenticate, redirecting to
register Form"
            value: "/login"
    /users/{id}:
      get:
        operationId: R106
        summary: "R106: View profile"
        description: "Showcase accounty features through personal profile.
Access: USR, MOD, ADM"
        tags:

```

```

- "M01: Account creation and accessibility"
parameters:
- in: path
  name: id
  schema:
    type: integer
    required: true
responses:
  "200":
    description: "Ok. Show User profile."

```

A8: Vertical Prototype

Artifact that seeks to implement the functionalities that define our application and its purpose.

1. Implemented User Stories

Identifier	Name	Priority	Description
US02	Over Text Search	Prototype	As a User, I want to have a text search correspondence bar, so that I can find the questions I am looking for.
US03	See Featured Questions	Prototype	As a User, I want to view the top questions so that, I know which ones have received more engagement.
US04	Browse Through Questions	Prototype	As a User, I want to go through the questions so that, I see the ones I like the most.
US13	Login (DTBMS)	Prototype	As a Guest, I want to log into the system using my website credentials, so that I can access all my account-related information.

US14	Register (DTBMS)	Prototype	As a Guest, I want to register myself on the BountyBust website, so that I can access it whenever I want to get onto the system.
US15	Logout	Prototype	As a Guest, I want to logout of my account, so that no one without my credentials can access my account on that device.
US18	Post Questions	Prototype	As an Authenticated User, I want to be able to post questions, so that other people can help me solve my problems.
US19	Answer Questions	Prototype	As an Authenticated User, I want to be able to answer other people's questions, so that I can share my expertise with others.
US21	View Profile	Prototype	As an Authenticated User, I want to view my profile, so that I have a unique area where my personal information is displayed.
US35	Content Edition	Prototype	As an Owner, I want to change the content of my questions or answers, so that I can update them if needed.
US36	Content Deletion	Prototype	As an Owner, I want to be able to delete my questions or answers, so that they are no longer available for other users.

Note: US35 and US36 were implemented partially only on questions.

2. Implemented Web Resources

- M01: Account creation and accessibility

Web Resource Reference	URL
R101: Login Form	GET /login
R102: Login Action	POST /login
R103: Logout Action	POST /logout
R104: Register Form	GET /register
R105: Register Action	POST /register
R106: View profile	GET /users/{id}

- M02: Bounties

Web Resource Reference	URL
R201: List of Bounties	GET /bounties
R202: Create a new bounty	POST /bounties/create
R203: Getting bounty create Form	GET /bounties/create
R204: Bounty Details	GET /bounties/{id}
R205: Delete Bounty	DELETE /bounties/{id}
R206: Edit Bounty Form	GET /bounties/{id}/edit
R207: Edit Bounty Action	POST /bounties/{id}/edit

- M05: Tags

Web Resource Reference	URL
R501: List of Tags	GET /tags

3. Prototype

The prototype Docker image is available at GitLab's Container Registry and can be run with:

```
docker run -d --name lbaw25146 -p 8001:80
gitlab.up.pt:5050/lbaw/lbaw2526/lbaw25146
```

You can run the BountyBust website at <http://localhost:8001>

Credentials:

- alice@example.com / hashed_pw_1

The submitted version of the code is available at <https://gitlab.up.pt/lbaw/lbaw2526/lbaw25146>

Revision History

Team

- José Miguel Ferreira Archer, up202307614@up.pt
- Marcelo Augusto Correia Lima, up202305892@up.pt